DP Biology – A1.1

Water

- Which features of water make life possible?
- What does living in water make easier and harder for organisms?

SL statements A1.1.1–6. Textbook pp. 2–10.
Image: Earth from Apollo 17. NASA, public domain.



Objectives for each class

Class	Statements	By the end of the class, you can...
1	A1.1.1–4	Explain how polar covalent bonds lead to hydrogen bonds, and how cohesion and adhesion matter to organisms
2	A1.1.5–6	Explain why water is a good solvent, and compare the physical properties of water and air for animals

The big picture: one property explains the rest



A water molecule is small and simple, but the electrons in it are shared unequally. That makes each molecule slightly charged at its ends, so neighbouring molecules attract each other by **hydrogen bonds**.

Those attractions explain why water sticks to itself and to other surfaces, why so many substances dissolve in it, and why it takes so much energy to heat. Each of these matters to living things.

Big question for this chapter

Why do scientists who search for life on other planets look for liquid water first?

Water is the medium for life

A1.1.1

The **first cells** arose in water. A membrane enclosed a small volume of water, and the reactions of life happened inside it.

Most of the processes of life still happen in water. It makes up most of the **cytoplasm**, it surrounds cells as **tissue fluid**, and it is the main part of **blood plasma**, xylem sap and phloem sap.

Because so much of an organism is water, the properties of water affect everything cells do.

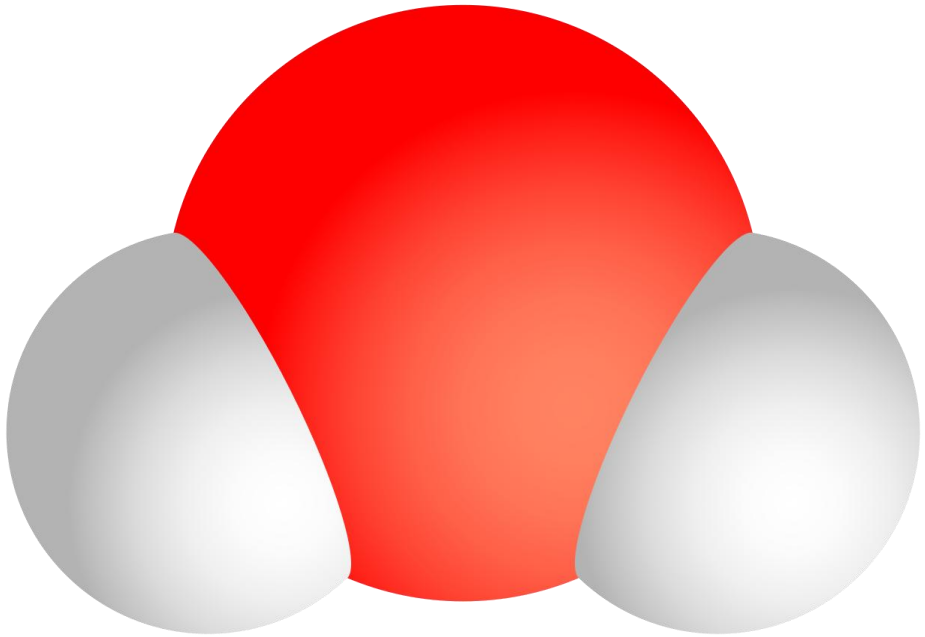
Organism or part	Water content
Human body	About 60%
Cytoplasm of a cell	About 70–80%
Lettuce leaf	About 95%
Jellyfish	About 95%
Dry seed	About 10%

Think about it

A dry seed is alive but almost inactive. Suggest why it cannot germinate until it takes up water.

Polar covalent bonds in a water molecule

A1.1.2



A water molecule. Sakurambo, public domain.

In a water molecule, an oxygen atom shares a pair of electrons with each of two hydrogen atoms. These are **covalent bonds**.

The sharing is **unequal**. The oxygen nucleus has more protons and pulls the shared electrons towards itself, so the oxygen end is slightly negative and each hydrogen is slightly positive.

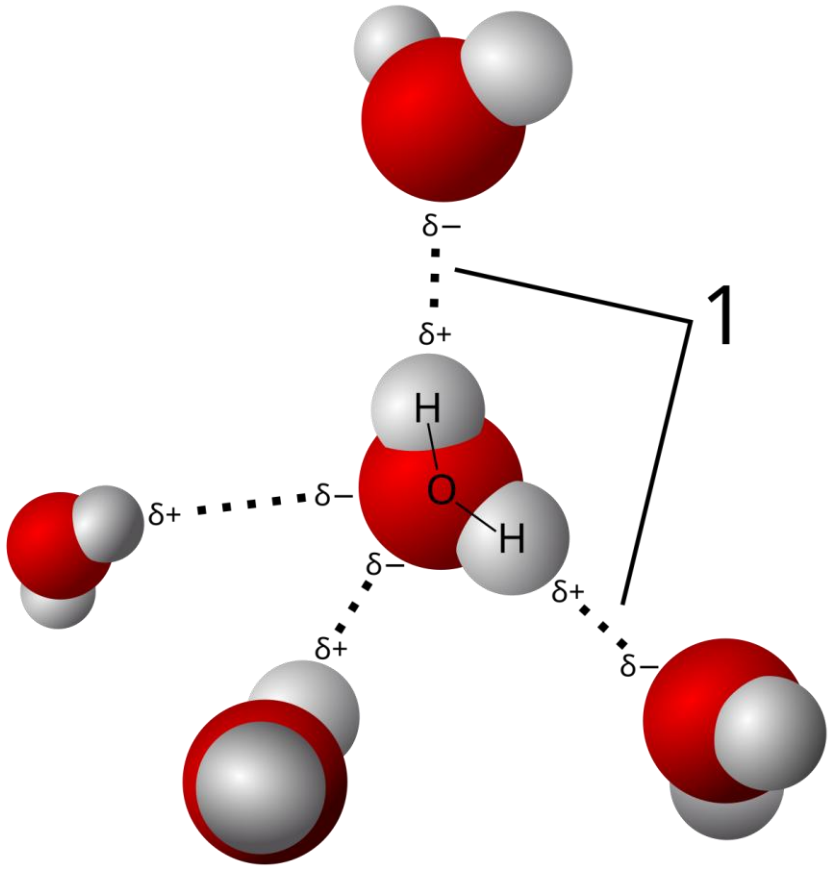
A bond with unequal sharing is a **polar covalent bond**. Because the molecule is bent, the charges do not cancel out, and the whole molecule is **polar**.

Notation

Show the slight charges with the Greek letter delta, as in the diagram on the next slides: delta-positive on each hydrogen, delta-negative on the oxygen.

Hydrogen bonds between water molecules

A1.1.2



Hydrogen bonds in water. Qwerter, public domain.

The slightly positive hydrogen of one molecule is attracted to the slightly negative oxygen of a neighbouring molecule. This attraction is a **hydrogen bond**.

A hydrogen bond is much weaker than a covalent bond, and in liquid water they constantly break and re-form. But there are so many of them that, together, they give water its unusual properties.

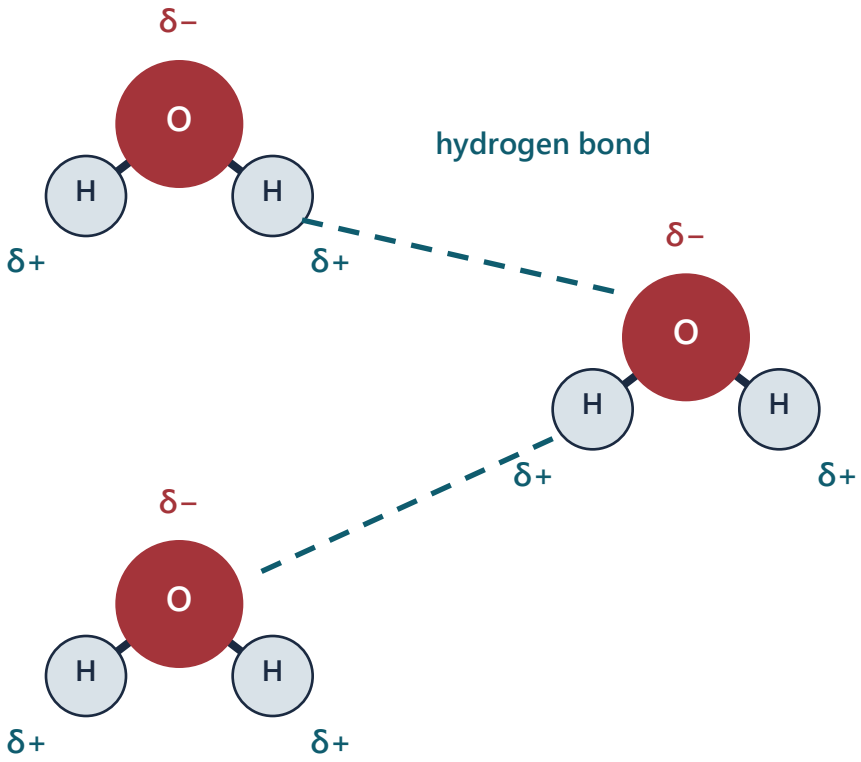
Each water molecule can form up to four hydrogen bonds.

Common mistake

A hydrogen bond is between molecules, not inside one. The bonds inside a water molecule are covalent.

How to draw water molecules and hydrogen bonds

A1.1.2



To score full marks in a drawing:

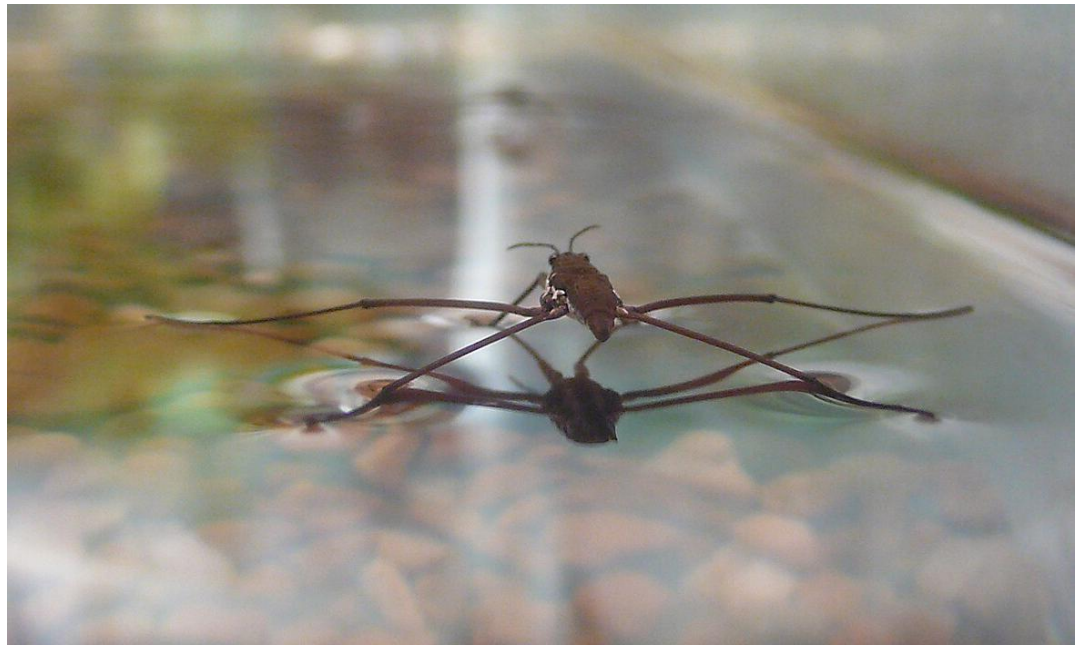
1. Draw each molecule as an O with two H atoms in a **bent** shape.
2. Label δ− on each oxygen and δ+ on each hydrogen.
3. Draw each hydrogen bond as a **dashed line** from an H of one molecule to the O of another.
4. Label at least one hydrogen bond, and use solid lines for the covalent bonds.

Try it

Draw four water molecules with at least three hydrogen bonds, then swap with a partner and check each other against the four points.

Cohesion: water molecules stick together

A1.1.3



A water strider on the surface. TimVickers, public domain.

Cohesion is the attraction of water molecules to each other, caused by hydrogen bonds.

At the surface, molecules are pulled inwards and sideways by their neighbours, so the surface behaves like a stretched skin. This is **surface tension**.

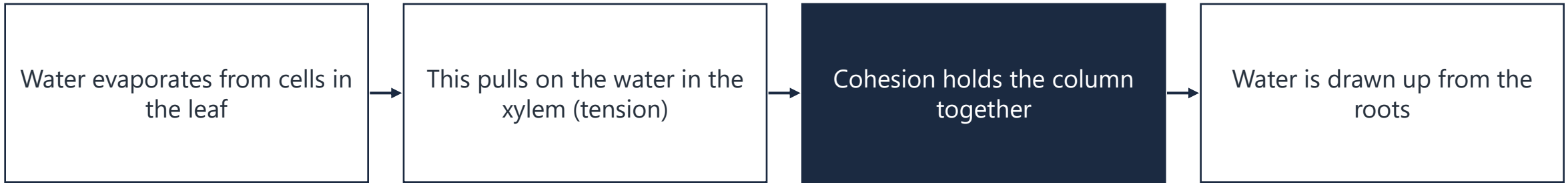
Small animals such as water striders can stand on the surface without breaking through, and use it as a habitat to hunt insects that fall in.

Key idea

Cohesion is strong enough to resist a pulling force. That is why a thin column of water can be pulled up a plant without breaking: the subject of the next slide.

Cohesion lets plants pull water up

A1.1.3



Xylem vessels are long, narrow, dead tubes running from the roots to the leaves. When water evaporates from the leaves, it pulls on the water below it, so the water in the xylem is under **tension**.

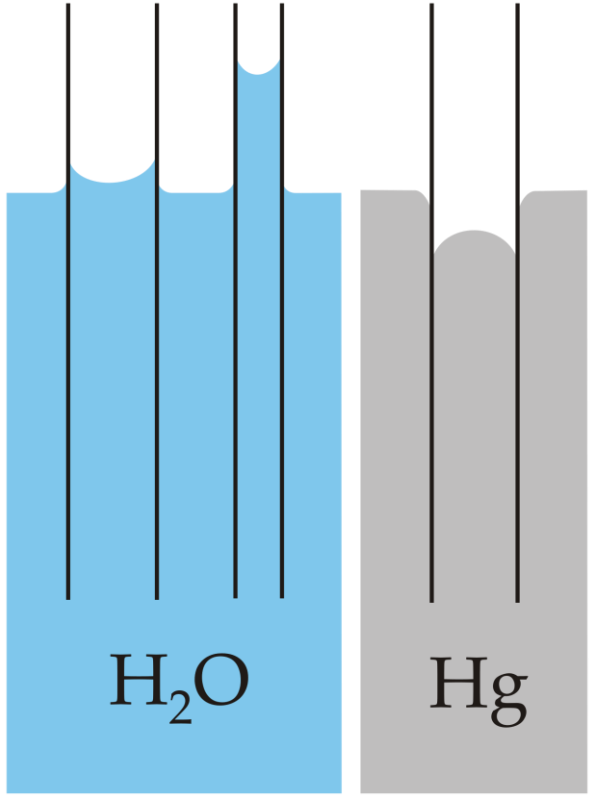
Because of cohesion, the pull is passed all the way down the column, and the column does not break. Water can be pulled to the top of trees over 100 m tall.

Link

B3.2 Transport in plants: transpiration and the structure of xylem vessels. Adhesion of water to the xylem walls also helps.

Adhesion: water sticks to other surfaces

A1.1.4



Capillary action in narrow tubes. MesserWoland, CC BY-SA 3.0.

Adhesion is the attraction of water to other substances whose surfaces carry charges or are polar, such as cellulose, glass or clay. Water forms hydrogen bonds with them.

Adhesion pulls water up the sides of a narrow tube, and cohesion pulls the rest of the water along behind it. Together they cause **capillary action**.

Where	What capillary action does
Soil	Draws water up from wetter soil below, through narrow spaces between particles, towards roots
Plant cell walls	Draws water through the gaps between cellulose fibres, so the walls of leaf cells stay wet

Common mistake

Cohesion: water to water. Adhesion: water to something else. Always say which.

Check-in: bonds, cohesion and adhesion

A1.1.1–4

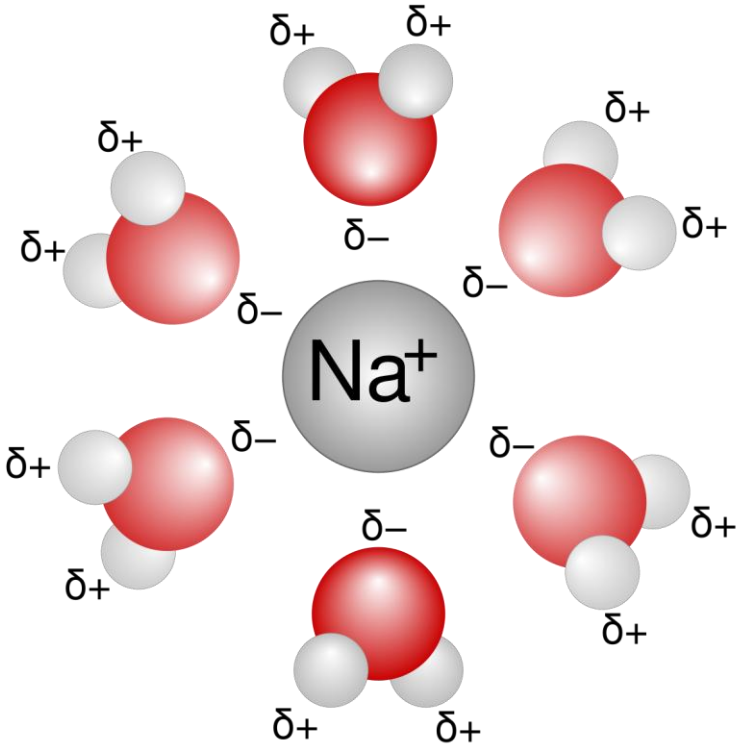
1. Explain why a water molecule is polar.
2. State the difference between a covalent bond and a hydrogen bond in water.
3. Draw two water molecules joined by a hydrogen bond, showing polarity.
4. Explain how cohesion allows water to move up a tall tree.
5. State one example of adhesion that matters to plants.

How to use this

Answer without your notes, then compare with a partner.

Water as a solvent

A1.1.5



Water surrounding a sodium ion. Steven P. Berg, public domain.

Because water is polar, it is attracted to **ions** and to other **polar molecules**. Water molecules cluster around each particle, separating it from the others, so it dissolves.

Around a positive ion such as Na^+, the δ− oxygens face inwards; around a negative chloride ion, the δ+ hydrogens face inwards.

Substances that dissolve easily in water are **hydrophilic** (water-loving): salts, sugars, amino acids and many proteins.

Why it matters

Water can dissolve a huge range of substances, so it can carry them around organisms and bring them together for reactions.

Water is the medium for metabolism

A1.1.5

Most reactions in cells happen in **aqueous solution**, in the cytoplasm or inside organelles. Dissolved molecules move freely and collide with each other and with enzymes.

Most **enzymes** catalyse reactions in solution, and their shapes depend on the water around them.

Water is also a reactant or product in many reactions: it is used in **hydrolysis** (breaking polymers down) and released in **condensation** (building them up), in photosynthesis and in respiration.

Reaction	Role of water
Digestion of starch	Used (hydrolysis)
Making a protein	Released (condensation)
Photosynthesis	Used, split to O_2
Aerobic respiration	Released

Link

B1.1 and B1.2: condensation and hydrolysis of carbohydrates, lipids and proteins.

How blood carries substances that do and do not dissolve

A1.1.5

Substance	In water	How blood carries it
Sodium chloride	Soluble	Dissolved in plasma as sodium and chloride ions
Glucose	Soluble	Dissolved in plasma: its -OH groups form hydrogen bonds with water
Amino acids	Soluble	Dissolved in plasma
Oxygen	Poorly soluble	Mostly bound to haemoglobin inside red blood cells
Fats and cholesterol	Insoluble	Inside lipoprotein particles, coated with phospholipids and proteins

Key idea

Blood can carry insoluble substances only by packaging them or binding them to proteins.

When being insoluble is useful

A1.1.5



Water droplets on a waxy leaf. Siddharth Patil, CC0.

Hydrophobic (water-fearing) molecules are non-polar, so they cannot form hydrogen bonds with water and do not dissolve. Many molecules work **because** they are hydrophobic.

The hydrophobic core of a **membrane** stops ions and polar molecules leaking through. **Fats** store energy without affecting water balance. A waxy **cuticle** on leaves reduces water loss.

Common mistake

Water does not repel hydrophobic molecules. Water molecules are more attracted to each other than to them, so the hydrophobic molecules end up pushed together.

Physical properties: water compared with air

A1.1.6

Property	What it means	Water compared with air	Consequence
Buoyancy	Upward force from the fluid; depends on its density	Water is about 800 times denser	Water supports the body
Viscosity	How strongly the fluid resists flow	About 50 times higher	More drag when moving
Thermal conductivity	How fast heat passes through	About 24 times higher	Body heat is lost fast
Specific heat	Energy needed to warm 1 g by 1 °C	About 4 times higher (4.2 J g^{-1} $°C^{-1}$)	Temperature changes slowly

Key idea

Water's high specific heat comes from hydrogen bonds: energy is used to break them before the molecules can move faster.

What these properties mean for animals in water

A1.1.6

Property	Challenge or opportunity	Example of an adaptation
Buoyancy	Water supports the body, so large animals can live in it; but staying down or up costs energy	Swim bladder in bony fish; fat or oil to float
Viscosity	Moving through water causes drag	Streamlined, torpedo-shaped bodies
Thermal conductivity	Warm-blooded animals lose heat quickly	Blubber in seals and whales; waterproof feathers
Specific heat	Water temperature is stable, so habitats change slowly	Many aquatic species tolerate only a narrow range

Link

B3.3 HL: adaptations of marine mammals for swimming.

Living in both water and air: the seal and the loon

A1.1.6



Ringed seal pup. Shawn Dahle, NOAA, public domain.



Black-throated loon. Robert Bergman, public domain.

	Ringed seal (Pusa hispida)	Black-throated loon (Gavia arctica)
Buoyancy	Blubber is less dense than water, helping it float	Denser bones than most birds, and it squeezes air from its feathers to dive
Viscosity	Streamlined body; flippers	Torpedo-shaped body; legs set far back with webbed feet
Heat loss	Thick blubber insulates in icy water	Dense, oiled, waterproof feathers trap air
In air	Moves slowly on ice; must surface to breathe	Must fly, so it cannot be too heavy; awkward on land

Key idea

Animals that use both media need compromises: what helps in water can be a problem in air.

Working with data: losing heat in water and in air

A1.1.6

Students filled flasks with water at 40 °C as model animals and recorded the temperature after 20 minutes. Some flasks were wrapped in a 1 cm layer of lard, as a model of blubber. These data are invented for practice; the pattern is realistic.

Flask	Start / °C	After 20 min / °C
In air at 10 °C	40	33
In water at 10 °C	40	21
In water, wrapped in lard	40	31

1. Calculate the temperature fall in each flask.
2. Explain why the flask in water cooled faster than the flask in air.
3. Explain the effect of the lard.
4. Suggest one way to improve the method.

Common mistakes to avoid

Not this	But this
Hydrogen bonds hold the atoms of a water molecule together.	Covalent bonds hold the atoms together. Hydrogen bonds are between molecules.
Water is polar because it contains oxygen.	It is polar because electrons are shared unequally and the molecule is bent.
Cohesion and adhesion are the same thing.	Cohesion: water to water. Adhesion: water to another polar or charged material.
Water repels hydrophobic molecules.	Water molecules attract each other more strongly, so the hydrophobic molecules are excluded.
Water heats up slowly because it is dense.	It heats slowly because energy is used to break hydrogen bonds (high specific heat).

Check that you can...

1. **Outline** why water is the medium for life
2. **Explain** why water molecules are polar, and **draw** hydrogen bonds between them
3. **Explain** cohesion, with xylem and surface tension as examples
4. **Explain** adhesion, with capillary action in soil and cell walls as examples
5. **Explain** why water is a good solvent, and how this links to metabolism and transport
6. **Outline** why some molecules work because they are hydrophobic
7. **Compare** the buoyancy, viscosity, thermal conductivity and specific heat of water and air
8. **Discuss** the adaptations of the ringed seal and black-throated loon

Exit activity: 5 minutes

On your own and without notes. Hand this in as you leave.

1. Explain why water molecules are polar.
2. Distinguish between cohesion and adhesion, giving one example of each in organisms.
3. Explain why a seal needs a thick layer of blubber in cold water.

Image credits

The Earth seen from Apollo 17.jpg – NASA/Apollo 17 crew; taken by either Harrison Schmitt or Ron – Public domain

Water molecule.svg – Sakurambo – Public domain

3D model hydrogen bonds in water.svg – User Qwerter at Czech wikipedia: Qwerter. Transferred from c – Public domain

Water strider.jpg – TimVickers – Public domain

Capillarity.svg – MesserWoland – CC BY-SA 3.0

Na+H2O.svg – Original: Dr. Steven P. Berg Vector: Duncan Keall, Glrx, JoK – Public domain

Water droplets on leaf.jpg – Siddharth Patil – CC0

Pusa hispida pup.jpg – Shawn Dahle, NOAA, Polar Ecosystems Program research cruise – Public domain

Gavia arctica1.jpg – Robert Bergman – Public domain

All from Wikimedia Commons (commons.wikimedia.org, search the file name). Some images are cropped. CC licences: creativecommons.org/licenses